

Retail Spatial Intelligence

CCTV Foot-Traffic & Spatial Behavior Report

VERIFIED COMPUTER VISION DATA

Analysis Parameters & Metadata

VIDEO SOURCE:	sample_retail_cctv.mp4	DATE PROCESSED:	Oct 6, 2026, 09:32 PM
RESOLUTION:	768x432 (768x432)	VIDEO DURATION:	49.67 seconds (596 frames @ 12 fps)
ZONE CONFIG:	Automatic Camera Regions	REPORT VERSION:	v2.1 (Canonical Analytics v2.0)
TRACKING QUALITY:	Good (High Confidence)	ANALYSIS ID:	d622a9e2-dbef-4ff8

UNIQUE VISITORS

7

Valid tracked customers

AVG ZONE DWELL

3.42s

Median: 4.5s

PEAK OCCUPANCY

3

Concurrent shoppers

TOTAL ZONE VISITS

9

Across 4 zones

Executive Summary

DETERMINISTIC SPATIAL RULES SUMMARY

Video analysis observed 7 unique tracked customer visits across 49.67s of footage. Overall average dwell time was 3.42s with a peak simultaneous store occupancy of 3 people. 2 potential bottleneck area(s) and 2 potential dead-zone area(s) were deterministically identified.

1. Computer Vision Processing & Traffic Diagnostics

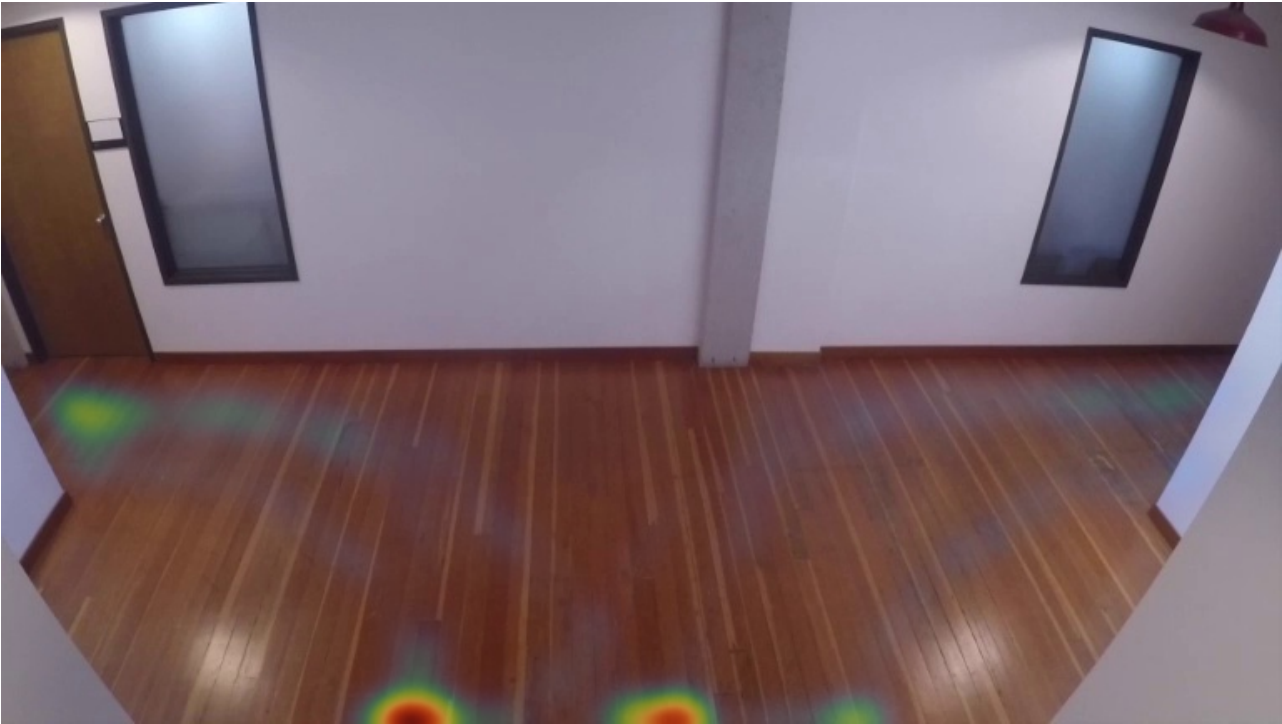
Footage was processed using Ultralytics YOLOv8 for human detection and ByteTrack for continuous anonymous multi-object tracking. Detection ground points were extracted at the bottom-center of bounding boxes ($x = [x1+x2]/2$, $y = y2$) to represent true floor contact. Tracks below the duration threshold of 1s were filtered to remove transient occlusion artifacts.

PROCESSING TELEMETRY

Raw Tracks Detected: 7	Effective Tracking Confidence: Good	Avg Detection Confidence: 0.83
Filtered Transient Tracks: 0	Average Track Duration: 4.5s	Relative Speed: 99 px/s

2. Observed Foot-Traffic Density Heatmap

Heatmap generated via 2D Gaussian Kernel density accumulation of verified customer foot-point coordinates overlaid onto the representative video frame.



Heatmap Methodology & Legend:

Blue = Low Pedestrian Activity | Green/Yellow = Moderate Dwell & Flow | Red/Crimson = Peak Traffic & Congestion Stagnation

3. Spatial Zone Performance Analysis

Deterministic measurements per defined spatial zone. Dwell sessions include a 0.8s occlusion buffer to prevent transient drops from breaking sessions.

Zone Name	Visits	Unique	Avg Dwell	Max Dwell	Peak Occ	Avg Occ	Share (%)	Density
Region 1 (Upper Left)	0	0	0s	0s	0	0	0%	0
Region 2 (Upper Right)	0	0	0s	0s	0	0	0%	0
Region 3 (Lower Left)	5	5	3.5s	4.83s	2	0.36	55.6%	6.03
Region 4 (Lower Right)	4	5	3.33s	4.83s	2	0.29	44.4%	6.03

4. Customer Flow & Traversal Pathways

Zone-to-zone transitions mined from consecutive spatial crossings across verified customer paths:

- DETECTED DIRECT ZONE TRANSITIONS:
- Region 4 (Lower Right) → Region 3 (Lower Left) : 2 visitor(s) (66.7%)
 - Region 3 (Lower Left) → Region 4 (Lower Right) : 1 visitor(s) (33.3%)

Frequent Customer Journey Sequences:

- Region 4 (Lower Right) → Region 3 (Lower Left)**
Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 4.2s
- Region 3 (Lower Left)**
Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 4.7s
- Region 4 (Lower Right)**
Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 4.7s
- Region 3 (Lower Left) → Region 4 (Lower Right)**
Observed in 1 customer(s) (14.3% of journeys) - Avg Duration: 4.3s

5. Forensic Bottleneck & Dead-Zone Intelligence

POTENTIAL BOTTLENECK ANALYSIS:

- **Region 3 (Lower Left) (Bottleneck Score: 80.4/100 - Confidence: High)**
 - Evidence: Elevated peak concurrency: 2 simultaneous customer tracks observed
 - Evidence: Prolonged dwell duration: 3.5s exceeds store average (3.42s)
 - Evidence: Movement slowdown: 96.8 px/s (2% slower than store baseline 99.0 px/s)
 - Evidence: High spatial traffic density: 6.03 visitors per 100k sq px
- **Region 4 (Lower Right) (Bottleneck Score: 78.5/100 - Confidence: Medium)**
 - Evidence: Elevated peak concurrency: 2 simultaneous customer tracks observed
 - Evidence: High spatial traffic density: 6.03 visitors per 100k sq px

POTENTIAL DEAD-ZONE ANALYSIS:

- **Region 1 (Upper Left) (Deficiency Score: 100/100, Activity: 0% - Confidence: High)**
 - Evidence: Zero customer visits recorded during observed footage
 - Evidence: Zero customer penetration observed
 - Evidence: Negligible concurrency: maximum 0 customer present simultaneously
- **Region 2 (Upper Right) (Deficiency Score: 100/100, Activity: 0% - Confidence: High)**
 - Evidence: Zero customer visits recorded during observed footage
 - Evidence: Zero customer penetration observed
 - Evidence: Negligible concurrency: maximum 0 customer present simultaneously

HIGH-ACTIVITY DESTINATION ZONES:

- Region 3 (Lower Left): 5 visits, 3.5s avg dwell, peak concurrent: 2 (High Activity Corridor)
- Region 4 (Lower Right): 4 visits, 3.33s avg dwell, peak concurrent: 2 (High Activity Corridor)

6. Strategic Recommendations & Priority Action Plan

[Congestion Alleviation] Region 3 (Lower Left)

Evidence: Elevated peak concurrency: 2 simultaneous customer tracks observed; Prolonged dwell duration: 3.5s exceeds store average (3.42s); Movement slowdown: 96.8 px/s (2% slower than store baseline 99.0 px/s); High spatial traffic density: 6.03 visitors per 100k sq px
Action: Widen primary aisle clearance or reposition high-interaction displays in Region 3 (Lower Left) to alleviate observed transit bottlenecks.

[Foot-Traffic Revitalization] Region 1 (Upper Left)

Evidence: Zero customer visits recorded during observed footage; Zero customer penetration observed; Negligible concurrency: maximum 0 customer present simultaneously
Action: Introduce focal visual merchandising, promotional signage, or destination product categories in Region 1 (Upper Left) to redistribute foot-traffic from congested zones.

Prioritized Action Plan:

- [High Priority]:** Audit aisle clearance in Region 3 (Lower Left), Region 4 (Lower Right) to eliminate movement slowdowns.
- [Medium Priority]:** Review product placement and category visibility in Region 1 (Upper Left), Region 2 (Upper Right).
- [Low Priority]:** Conduct subsequent video analysis after implementing spatial adjustments to evaluate before/after metrics.

7. Scientific Methodology & Audit Traceability

- YOLOv8 Person Detection**
Pretrained neural model detecting COCO Class 0 (person). Frame sampling optimizes CPU throughput without altering physical timeline.
- ByteTrack Multi-Object Tracking**
Associates detections using high/low confidence thresholds and Kalman state estimation with persistent anonymous IDs (Customer #1, #2).
- Floor Foot-Point Extraction**
Computes ground contact coordinate at $x = (x1+x2)/2$, $y = y2$ to bypass perspective bounding box tilt.
- Point-in-Polygon Zone Assignment**
Assigns floor foot-points to arbitrary user-drawn polygons or balanced camera regions using OpenCV pointPolygonTest.
- Dwell Time with Occlusion Grace Buffer**
Maintains an 0.8s tolerance buffer before closing dwell sessions to prevent brief tracking drops from artificially splitting sessions.
- 2D Gaussian Density Heatmaps**
Accumulates foot-points into a spatial matrix smoothed with a normalized Gaussian kernel and mapped through OpenCV Turbo colormap.

8. System Limitations & Forensic Transparency

- Single-Camera Monocular Perspective:** Field of view is restricted to the camera mounting angle. Speeds are reported in image-space pixels/second, not calibrated metric coordinates.
- Visual Occlusion:** Dense crowds or structural pillars can temporarily occlude foot points. The platform uses a 0.8s tolerance buffer, but long occlusions may cause track fragmentation.
- Zero PII Policy:** The system operates with transient anonymous IDs only. No face recognition, demographic profiling, or biometric storage is performed.
- No POS Integration:** Analysis reflects physical pedestrian movement only. Purchase conversions, basket size, and revenue metrics require external POS pairing.

AUTHENTICATION & VERIFICATION STAMP

Generated deterministically by Retail Spatial Intelligence Platform Engine v2.1. All metrics directly traceable to video frame detections. Job ID: d622a9e2-dbef-4ff8-a719-6c5aeac08078

